Voltage: energy per charge

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Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Interpret potential difference as energy transferred per unit charge.

SUCCESS CRITERIA - I can...

1. Use volts as joules per coulomb.
2. Calculate energy transfer with E=qV.
3. Distinguish voltage from current.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “voltage” and “charge” to explain your answer. Sentence starters are optional.

voltage: electric potential difference; energy transferred per charge

charge: a property responsible for electric interactions

energy: a quantity tracked through storage and transfers in a system

01 Warm-up: bring back an earlier idea

A neutral object is attracted to an object carrying **+3 nC**. **Explain how attraction can occur** without giving the neutral object a net charge.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: energy transferred = qV; V = energy transferred/q. Magnitudes here; **1 V** = **1 J/C**. Voltage is a difference between two points.

Worked reasoning: Moving **2 C** through a potential difference of **3 V** changes electric potential energy by q ΔV=**6 J**. Voltage is energy change per charge, not energy by itself.

Your turn: Draw two labeled potential levels and an energy account for a positive charge moved between them.

Sentence starter: In my work, I used … to show … . I labeled it … .

Initial / supply _____ Final / component _____

V: ____ J/C V: ____ J/C

Charge: ____ C Energy transfer: ____ J

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Compare a 3 V** and a **6 V** battery for the same amount of charge transferred.
2. Which supplies more energy per coulomb?

Sentence starter: Both … . They differ in … because … .

Initial / supply _____ Final / component _____

V: ____ J/C V: ____ J/C

Charge: ____ C Energy transfer: ____ J

Marzano check - circle: ① ② ③

04 Apply the physics

A battery supplies **240 J** while **20 C** passes through it. **Find its potential difference**. In a separate component, **5 C** passes through a **3 V** drop. **Calculate the energy transferred there** and **label the two cases**.

Sentence starter: Both … . They differ in … because … .

Initial / supply _____ Final / component _____

V: ____ J/C V: ____ J/C

Charge: ____ C Energy transfer: ____ J

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A student says a **9 V** battery contains **9** coulombs. **Explain the units** and why voltage does not state total battery capacity.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

Sensors **1.36 m** apart record delay **0.004 s**. **Find wave speed**. If **0.001 s** is added to the delay, find the incorrect speed and **describe the bias**.
Label the full travel path and the recorded time.

Sentence starter: The step I would change is … . I would change it because I found … .



Marzano check - circle: ① ② ③

07 Exit check

Two components in series have drops **2 V** and **4 V**. For **3 C** passing through each, find each energy transfer and the total.

Sentence starter: I found … [unit]. This value means … in this situation.

Initial / supply	Final / component
V: ____ J/C	V: ____ J/C

Charge: ____ C Energy transfer: ____ J

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A charge **-2** microcoulombs moves through ΔV=**+50 V**.
1. Find ΔU and work by the electric field.
2. **Explain both signs**.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Initial / supply	Final / component
V: ____ J/C	V: ____ J/C

Charge: ____ C Energy transfer: ____ J

Marzano check - circle: ① ② ③